Status	Finished
Started	Saturday, 17 October 2026, 9:25 PM
Completed	Saturday, 17 October 2026, 9:31 PM
Duration	5 mins 59 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. There should be direct access to the middle element in any sublist

c. There must be a mechanism to delete and/or insert elements in the list ✓

d. Number values should only be present

Number values should only be present

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓
b. The search continues in the right sublist
c. The list is sorted
d. The search continues in the left sublist

Question **3**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search
b. Binary search
c. Linear search ✓
d. Hashing

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 89 and 94
c. 90 and 94
d. 90 and 99 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search requires the list to be sorted.
c. Linear search is more efficient than binary search on large lists.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and unsorted ✓
b. When the list is large and sorted
c. When the list is large and unsorted
d. When the list is small and sorted

Question **7**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. When the item is somewhere in the middle of the array ✓
c. When the item is the last element in the array
d. Item is the last element in the array or item is not there at all

Question **8**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. In the right sublist ✓
c. In the middle of the list
d. At the beginning of the list

Question **9**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. 1
c. n ✓
d. n/2

Question **10**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Linear search
c. Hash search
d. Binary search ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first

b. By comparing each element in the list sequentially ✓

c. By dividing the list into halves

d. By using a hash function

Question **12**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)

b. Hash

c. Linear

d. Binary ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially

b. By starting from the first element

c. By using a hash function

d. By dividing the list length by two ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers

b. It requires the list to be sorted ✓

c. It can only be applied to large lists

d. It is slower than linear search for small lists

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)

b. O(n log n)

c. O(log n)

d. O👎 ✓